

Full symmetric monodromy in a uniform Alpage-Gallagher subfamily

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Alec Kriebel, with heavy assistance from ChatGPT 5.6 Sol

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ABSTRACT

Gallagher's weighted-lift construction, posted after Alpage's announced counterexample, produces a three-dimensional Keller map of every generic degree $n \geq 3$. We isolate a simple uniform subfamily and prove that its function-field extension has full symmetric Galois closure S_n , hence no nonidentity rational deck transformations. A uniform rational collision is also given. This is a structural refinement of the Alpage-Gallagher examples, not an independent construction of the underlying counterexamples.

1. The explicit family

Fix an integer $n \geq 3$ and put

$$u = 1 + xy, \quad \text{gamma} = 1 - [n/(n-1)]xy + x^2z.$$

Define $F_n = (A_n, B_n, C_n)$ by

$$\begin{aligned} A_n &= \{(n-2)u + u^2 - (n-1)u^n \text{gamma}^{n-2}\} / \{(n-2)x^2\}, \\ B_n &= \{(n-2) + 2u - n u^{n-1} \text{gamma}^{n-2}\} / \{(n-2)x\}, \quad C_n = x \text{gamma}. \end{aligned} \quad (1)$$

The apparent quotients by x^2 and x cancel identically, so (1) is a polynomial map over $\mathbb{Q}$.

Theorem 1. For every $n \geq 3$, the map F_n is polynomial and $\det JF_n = 1$; the extension $C(x,y,z)/C(F_n)$ has degree n and Galois closure S_n ; and every rational map σ satisfying $F_n \circ \sigma = F_n$ is the identity.

2. Polynomiality and constant Jacobian

Set $v = xy$, $\tau = x^2z$, and $w = u \text{gamma}$. Define

$$p_n(w) = [2w - n w^{n-1}]/(n-2), \quad q_n(w) = [w^2 - (n-1)w^n]/(n-2).$$

Then $q_n' = w p_n'$ and (1) is $(\alpha/x^2, \beta/x, x \text{gamma})$, where $\alpha = u + q_n(w)/\text{gamma}^2$ and $\beta = 1 + p_n(w)/\text{gamma}$. As polynomials in (v, τ) , β lies in (v, τ) and α lies in (v^2, τ) ; substitution proves the required divisibility.

Put $P = BC = \text{gamma} + p_n(w)$ and $Q = AC^2 = w \text{gamma} + q_n(w)$. The output change $(A, B, C) \rightarrow (P, Q, C)$ has Jacobian $-C^3$. The input changes through (x, w, gamma) and the identity $w p' - q' - \text{gamma} = -\text{gamma}$ give the same Jacobian $-C^3$. Therefore $\det JF_n = 1$.

3. Inverse equation and degree

With $R_n(w) = (w^2 - w^n)/(n-2)$, the identity $q_n = w p_n - R_n$ eliminates gamma and gives $R_n(w) = wP - Q$. Thus the fiber polynomial is

$$H_n(T) = T^n - T^2 + (n-2)BCT - (n-2)AC^2. \quad (2)$$

A root w with $\text{gamma} = P - p_n(w)$ nonzero recovers the source rationally: $x = C/\text{gamma}$, $u = w/\text{gamma}$, $y = (u-1)/x$, and $z = \{\text{gamma}-1 + [n/(n-1)](u-1)\}/x^2$.

Writing $U = (n-2)BC$ and $V = -(n-2)AC^2$, the target field is $K = C(A,B,C) = C(U,V,C) = C(U,V)(C)$, since $B = U/((n-2)C)$ and $A = -V/((n-2)C^2)$. Here C is transcendental over $C(U,V)$. Equation (2) is $h_n(T) = T^n - T^2 + UT + V$. It is irreducible over $C(U,V)$: being linear in V with coefficient one, any factor independent of V would be a unit. Gauss's lemma preserves irreducibility under the purely transcendental base change to K . Hence the extension degree is n .

4. Full symmetric monodromy

The discriminant curve of h_n is parametrized by

$$t \rightarrow (U,V) = (2t - n t^{n-1}, (n-1)t^n - t^2). \quad (3)$$

It is irreducible. At $t = 0$ the polynomial is $T^2(T^{n-2}-1)$, with exactly one double root. It is the unique parameter over $(U,V) = (0,0)$, and the derivative of (3) has first coordinate 2, so this is a smooth discriminant point. The one-double-root condition is nonempty and Zariski open on the irreducible curve, so generic inertia is a transposition. The root-incidence variety is A^2 , hence connected, so the monodromy action is transitive. Quotienting the Galois closure by the normal subgroup generated by finite inertia yields a cover of A^2 unramified in codimension one. Purity makes it finite etale, and A^2 has no nontrivial connected finite etale cover. Therefore the monodromy group is generated by conjugate transpositions. A transitive group generated by transpositions is S_n . Finally, a finite algebraic extension is linearly disjoint from the purely transcendental extension $C(U,V)(C)$, so this remains the Galois group over the actual target field K .

Corollary. The rational deck group is trivial. In the S_n -closure, the root field is fixed by a point stabilizer S_{n-1} , which is self-normalizing; hence $\text{Aut}_K(L) = N(S_{n-1})/S_{n-1} = 1$.

5. Uniform rational collision

Let $s_n = (4 - 2^n)/(n-2)$. At the target $(s_n, s_n, 1)$, the fiber polynomial is $T^n - T^2 + (4-2^n)(T-1)$, with simple roots 1 and 2. For r in $\{1,2\}$, set $g_r = s_n - p_n(r)$ and

$$X_{n,r} = (1/g_r, r-g_r, g_r^2[g_r-1 + n/(n-1)(r/g_r-1)]). \quad (4)$$

Here $g_1 = (n+2-2^n)/(n-2)$ and $g_2 = 2^{n-1}$, both nonzero. Exact substitution gives $F_n(X_{n,1}) = F_n(X_{n,2}) = (s_n, s_n, 1)$. For $n = 3$: $F_3(-1/3, 4, -54) = F_3(1/4, -2, 36) = (-4, -4, 1)$.

6. Status and reproducibility

The every-degree lift and cubic S_3 analysis predate this note. Its narrow residual claims are the subfamily (1), uniform collision, all-degree S_n theorem, and trivial deck group. Searches on 20 July 2026 found no prior statement, but establish no priority. The exact checker accompanies the note; expert review is required before submission.

References

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Disclosure: Alec Kriebel is a complete amateur and cannot independently verify the mathematical claims in this note. The formulas, proof organization, verifier, and typeset draft were developed with heavy assistance from ChatGPT 5.6 Sol. Independent expert review is required.